

AP STATISTICS · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 9 UNITS · THE 3 DISTRIBUTIONS

About this exam. AP Statistics is half a math course and half a writing course. Half the FRQ rubric isn't "is the math right" — it's "did you write the conclusion in correct statistical English with the proper conditional language?" The signature challenge: knowing **WHICH inference procedure to choose** from a 12-item menu, then writing the **template-perfect** conclusion.

THE MATH OF YOUR TARGET

Target	MCQ (40q, 50%)	FRQ (6q, 50%)	Approx %	Tier
3 (PASS)	~22 / 40 (~55%)	~14/30 raw	~46-50%	[3] only
4	~27 / 40 (~68%)	~19/30 raw	~58-62%	[3] + [4]
5	~33 / 40 (~83%)	~24/30 raw	~70-75%	All

Time: 90 min for 40 MCQ (~2:15 each), 90 min for 6 FRQ (~15 min each, but Q6 is a 25-min "investigative task"). The investigative task ALONE is ~25% of FRQ score.

THE 9 UNITS — by exam weight

#	Unit	Weight	Focus
1	Exploring One-Variable Data	15–23%	Mean, median, SD, z-scores , normal model, boxplots
2	Exploring Two-Variable Data	5–7%	Scatterplots, correlation r , LSRL, residuals
3	Collecting Data	12–15%	Sampling methods , experiments vs observational, bias
4	Probability & Random Variables	10–20%	Independence, binomial + geometric , expected value
5	Sampling Distributions	7–12%	The crucial bridge. Mean + SD of a sample statistic. CLT.
6	Inference: Proportions	12–15%	1-prop + 2-prop z-tests + confidence intervals
7	Inference: Means	10–18%	1-sample + 2-sample t-tests + confidence intervals; matched pairs
8	Chi-Square	2–5%	Goodness of fit , homogeneity, independence
9	Inference for Slopes	2–5%	t-test for slope (regression); often appears in Q6 investigative task

THE 3 DISTRIBUTIONS — never confuse them again [3]/[4]/[5]

Distribution	What it is
Population distribution [3]	Distribution of values in the WHOLE population. Usually unknown — what we want to know.
Distribution of the data (sample) [3]	Distribution of values in YOUR ONE sample. What you actually see in a histogram of your data.
Sampling distribution of a statistic [4]	Distribution of a STATISTIC (e.g., $\bar{x}$ or $\hat{p}$) over ALL possible samples of size n . Mean = parameter (μ or p). SD shrinks as n grows.

KEY INSIGHT [5] This distinction is the single most-tested concept on AP Stats FRQs. The **CENTER** of a sampling distribution is the population parameter. The **SD** of a sampling distribution shrinks like $1/\sqrt{n}$. Mix these up = lose 2-3 points instantly.

PAGE 2 | UNIT 1: EXPLORING ONE-VARIABLE DATA

Describing Distributions — "S O C S" template [3]

S – Shape (symmetric / skewed-right / skewed-left / uniform / bimodal)
 O – Outliers (any unusual values?)
 C – Center (mean if symmetric, MEDIAN if skewed)
 S – Spread (SD if symmetric, IQR if skewed)

ALWAYS describe shape FIRST – it dictates whether to use mean/SD or median/IQR.
 ALWAYS write in CONTEXT – name the variable, state units.

WATCH OUT [3] Most-missed FRQ point. Describing distribution without naming the variable. "The distribution is skewed right" earns 0. "The distribution of **commute times** is skewed right" earns the point.

Center & Spread Statistics [3]/[4]

Mean: $\bar{x} = \Sigma x / n$ (sensitive to outliers – "non-resistant")
 Median: middle value (resistant to outliers)
 Variance: $s^2 = \Sigma (x - \bar{x})^2 / (n-1)$
 Std Dev: $s = \sqrt{s^2}$ (units same as data)
 Range: max – min
 IQR: $Q3 - Q1$ (resistant)
 Outlier rule ($1.5 \times \text{IQR}$): below $Q1 - 1.5 \cdot \text{IQR}$ or above $Q3 + 1.5 \cdot \text{IQR}$

Z-Scores & The Normal Model [3]/[4]

Z-SCORE: $z = (x - \mu) / \sigma$ "How many SDs above/below the mean"

EMPIRICAL RULE (only for normal distributions):

~68% within $\mu \pm 1\sigma$
 ~95% within $\mu \pm 2\sigma$
 ~99.7% within $\mu \pm 3\sigma$

CALCULATOR: `normalcdf(low, high,  $\mu$ ,  $\sigma$ )` ← finds AREA between bounds
`invNorm(area,  $\mu$ ,  $\sigma$ )` ← finds X given AREA below

KEY INSIGHT [4] Z-scores ALLOW comparison across different distributions. "Sue scored 85 on a test with $\mu=70$, $\sigma=10 \rightarrow z=1.5$. Bob scored 90 on a test with $\mu=80$, $\sigma=8 \rightarrow z=1.25$. Sue did relatively better."

Boxplots, Histograms, Dotplots [3]

- **Boxplot [3]:** box from $Q1$ to $Q3$, line at median, whiskers to min/max (or fences). Shows shape and outliers fast.
- **Modified boxplot [4]** = whiskers stop at fences; outliers shown as dots.
- **Skew direction [3]:** boxplot's longer tail = direction of skew. Right-skewed = mean > median.

PAGE 3 | UNIT 2: TWO-VARIABLE DATA · LSRL · RESIDUALS

Correlation r [3]/[4]

Range: $-1 \leq r \leq 1$

r near +1 = strong positive linear
 r near -1 = strong negative linear
 r near 0 = NO LINEAR RELATIONSHIP (could still be curved!)

r is RESISTANT to scale change (units, multiplying)
 r is NOT RESISTANT to outliers – one bad point can swing it
 r has NO UNITS

WATCH OUT [3] $r=0$ does NOT mean "no relationship" — it means "no LINEAR relationship". A perfect parabola $y=x^2$ has $r=0$. Scatterplot first, r second.

Least-Squares Regression Line (LSRL) [3]/[4]

$\hat{y} = a + bx$ (READ: "y-hat = a plus b times x")

Slope: $b = r \cdot (s_y / s_x)$

Intercept: $a = \bar{y} - b \cdot \bar{x}$ (LSRL ALWAYS passes through $(\bar{x}, \bar{y})$)

r^2 = COEFFICIENT OF DETERMINATION

= proportion of variation in y EXPLAINED by linear regression on x

interpret as percent: " $r^2 = 0.81 \rightarrow 81\%$ of variation in y explained by x "

KEY INSIGHT [4] Slope interpretation template. "For each additional [unit of x], the predicted [y , in context] increases/decreases by [b] [units]." — must be in CONTEXT and **predicted** y , not actual.

Residuals & Residual Plots [3]/[4]

Residual = ACTUAL - PREDICTED = $y - \hat{y}$

POSITIVE residual $\rightarrow$ actual ABOVE the line $\rightarrow$ LSRL UNDERpredicted

NEGATIVE residual $\rightarrow$ actual BELOW the line $\rightarrow$ LSRL OVERpredicted

RESIDUAL PLOT (residual vs x):

NO PATTERN (random scatter) $\rightarrow$ linear model is APPROPRIATE ✓

CLEAR CURVED PATTERN $\rightarrow$ linear model is NOT appropriate (try transform)

WATCH OUT [4] Common FRQ trap. Asked "is a linear model appropriate?" The answer comes from the RESIDUAL plot, NOT from r . A high r can mask curvature.

Influential Points, Outliers, Leverage [4]/[5]

- **Outlier in regression [4]** = point with large residual.
- **Influential point [4]/[5]** = point whose REMOVAL substantially changes the LSRL slope/intercept (typically extreme x -values).
- **[5] High-leverage point** = far from $\bar{x}$. Influential $\leftrightarrow$ high leverage AND large residual usually.

Transformations [4]/[5]

- **[4] If residual plot curves**, try transforming: $\log(y)$, $\sqrt{y}$, $1/y$, $\log(x)$. Compare residual plots — pick the one that's most random.
- **[5] Common back-transforms**: if $\log(\hat{y}) = a + bx$, then $\hat{y} = 10^{(a+bx)}$. Be careful with units.

PAGE 4 | UNITS 3-4: DATA COLLECTION · PROBABILITY · RANDOM VARIABLES

Sampling Methods [3]/[4]

Method	How it works
SRS (Simple Random Sample) [3]	Every group of size n equally likely. Gold standard. Random number table or generator.
Stratified [3]/[4]	Population split into HOMOGENEOUS strata; SRS within each. Reduces variability.
Cluster [3]/[4]	Population split into HETEROGENEOUS clusters; randomly select WHOLE clusters. Convenient when SRS impractical.
Systematic [4]	Pick every k th person from an ordered list. Easy but vulnerable if there's a pattern in the list.
Convenience / Voluntary Response [3]	BAD. Self-selection bias, undercoverage. Reject these as "not random".

Bias [3]/[4]

- **Voluntary response bias** [3] = those who feel strongly self-select.
- **Undercoverage** [3]/[4] = part of population systematically excluded (phone surveys miss those without phones).
- **Nonresponse bias** [3]/[4] = selected respondents don't reply, may differ systematically.
- **Response bias** [3]/[4] = answers shaped by question wording, interviewer, sensitivity of topic.
- **[4] Random sampling reduces selection bias but cannot eliminate response/nonresponse bias.**

Experiments vs Observational Studies [3]/[4]

- **Observational study** [3]: just observe — CANNOT prove causation.
- **Experiment** [3]/[4]: imposes a TREATMENT. Random assignment + control + replication → CAN prove causation (within scope of study).
- **Blocking** [4] = group similar units into BLOCKS, randomize within. Reduces variability. "*Block what you CAN, randomize what you can't.*"
- **Matched pairs** [4] = special block of size 2. Each subject acts as own control (before/after) OR pairs of similar subjects.
- **Blinding** [3]/[4]: single-blind (subjects don't know), double-blind (neither subjects nor evaluators).

Probability Rules [3]/[4]

Complement: $P(\text{not } A) = 1 - P(A)$

Addition: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Mutually exclusive (disjoint): $P(A \text{ and } B) = 0 \rightarrow P(A \text{ or } B) = P(A) + P(B)$

Conditional: $P(A | B) = P(A \text{ and } B) / P(B)$

Multiplication: $P(A \text{ and } B) = P(A) \cdot P(B | A)$

Independent: $P(A \text{ and } B) = P(A) \cdot P(B)$

Independence test: $A \text{ and } B \text{ independent} \Leftrightarrow P(A | B) = P(A)$

Discrete Random Variables — Binomial vs Geometric [3]/[4]

BINOMIAL (FIXED number of trials, count successes):

Conditions: Binary, Independent, fixed N , equal P ("BINS")

Mean: $\mu = np$ Variance: $np(1-p)$

Calc: $\text{binompdf}(n, p, x)$ $\text{binomcdf}(n, p, x)$

GEOMETRIC (count trials UNTIL FIRST SUCCESS):

Same conditions except no fixed N

Mean: $\mu = 1/p$

Calc: $\text{geompdf}(p, x)$ $\text{geometcdf}(p, x)$

PAGE 5 | UNIT 5: SAMPLING DISTRIBUTIONS — THE BRIDGE TO INFERENCE

This unit is short but it's the conceptual heart of the course. Every inference procedure in Units 6-9 is built on the sampling distribution. Master this and the rest snaps into place.

Sampling Distribution of $\hat{p}$ (sample proportion) [3]/[4]

Center: $\mu_{\hat{p}} = p$ (UNBIASED estimator)
 Spread: $\sigma_{\hat{p}} = \sqrt{p(1-p) / n}$
 Shape: APPROX NORMAL if $np \geq 10$ AND $n(1-p) \geq 10$

CONDITIONS to use this normal approximation:

1. RANDOM sample (or random assignment)
2. INDEPENDENT – 10% rule: $n \leq 10\%$ of population
3. NORMAL – Large Counts: $np \geq 10$ AND $n(1-p) \geq 10$

Sampling Distribution of $\bar{x}$ (sample mean) [3]/[4]

Center: $\mu_{\bar{x}} = \mu$ (UNBIASED)
 Spread: $\sigma_{\bar{x}} = \sigma / \sqrt{n}$ (gets SMALLER as n grows!)

Shape:

If population is NORMAL $\rightarrow \bar{x}$ is normal for ANY n
 Otherwise $\rightarrow$ CENTRAL LIMIT THEOREM: $\bar{x}$ approx normal if $n \geq 30$

CONDITIONS to use this normal approximation:

1. RANDOM sample
2. INDEPENDENT – 10% rule
3. NORMAL – population normal OR $n \geq 30$ (CLT) OR sample looks ~symmetric

KEY INSIGHT [4] CLT in plain English. No matter what shape the population has, if you take a big enough sample ($n \geq 30$), the DISTRIBUTION OF SAMPLE MEANS will be approximately normal, centered at the true population mean, with $SD = \sigma/\sqrt{n}$. **This is what makes statistics work.**

Differences of Two Sample Statistics [4]/[5]

Difference of proportions $\hat{p}_1 - \hat{p}_2$:

Mean: $p_1 - p_2$

SD: $\sqrt{p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2}$

Difference of means $\bar{x}_1 - \bar{x}_2$:

Mean: $\mu_1 - \mu_2$

SD: $\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$ (variances ADD, not SDs!)

KEY: variances add, SDs do NOT add directly

Bias vs Variability [3]/[4]

- **Bias [3]** = sampling distribution centered AT WRONG VALUE (not at parameter). A biased sampling method consistently misses the target.
- **Variability [3]** = how spread out the sampling distribution is. **Larger $n \rightarrow$ smaller variability** (more precision).
- **[4]** Bias = the dartboard problem: arrows clustered but not at bullseye = biased. Arrows scattered around bullseye = unbiased but variable.

PAGE 6 | UNITS 6-7: INFERENCE — PROPORTIONS · MEANS · TEMPLATES

THE 4-STEP INFERENCE TEMPLATE — "PHANTOMS" [3]/[4]

P Parameter – define in CONTEXT (μ , p , $\mu_1 - \mu_2$, etc.)
 H Hypotheses – H_0 and H_a in symbols + words
 A Assumptions / Conditions – VERIFY each (random, 10%, large counts/normal)
 N Name the test (1-prop z, 2-sample t, χ^2 GOF, etc.)
 T Test statistic + p-value (show formula or calculator output)
 O Obtain conclusion in CONTEXT
 M Make decision (reject / fail to reject H_0 at α)
 S State conclusion in CONTEXT ("there IS / IS NOT convincing evidence...")

THE CONCLUSION TEMPLATES — memorize verbatim [3]/[4]

REJECT H_0 : "Since p-value (____) < α (____), we REJECT H_0 .
 There IS convincing evidence that [H_a in context]."
 FAIL TO REJECT: "Since p-value (____) > α (____), we FAIL to reject H_0 .
 There is NOT convincing evidence that [H_a in context]."
 CONFIDENCE INTERVAL: "We are ____% confident that the true [parameter
 in context] is between ____ and ____."

WATCH OUT [3] NEVER write "accept H_0 ". It's always "fail to reject" — you didn't prove H_0 , you just didn't gather enough evidence against it.
 "Accept" loses the conclusion point.

PROPORTION TESTS [3]/[4]

1-PROP Z-TEST (one population proportion):
 $H_0: p = p_0$ test stat: $z = (\hat{p} - p_0) / \sqrt{p_0(1-p_0)/n}$
 1-PROP Z-INTERVAL:
 $\hat{p} \pm z^* \cdot \sqrt{\hat{p}(1-\hat{p})/n}$ (use $\hat{p}$, not p_0 !)
 2-PROP Z-TEST (compare two populations):
 $H_0: p_1 = p_2 \rightarrow$ pooled $\hat{p}_c = (x_1 + x_2) / (n_1 + n_2)$
 test stat: $z = (\hat{p}_1 - \hat{p}_2) / \sqrt{\hat{p}_c(1-\hat{p}_c)(1/n_1 + 1/n_2)}$
 2-PROP Z-INTERVAL – uses SEPARATE $\hat{p}$ s, NOT pooled:
 $(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$

MEANS TESTS — "t", not z [3]/[4]

Use t-distribution (NOT z) when σ is UNKNOWN – virtually always in practice.

1-SAMPLE T-TEST: $df = n - 1$
 $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$
 1-SAMPLE T-INTERVAL:
 $\bar{x} \pm t^* \cdot (s/\sqrt{n})$
 2-SAMPLE T-TEST (independent samples, unpooled is the AP standard):
 $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
 $df =$ use calculator's value (Welch's approximation), or smaller of $n_1 - 1$, $n_2 - 1$
 MATCHED PAIRS T-TEST (paired data $\rightarrow$ take differences first, then 1-sample t)
 test stat on the differences: $t = (\bar{x}_d - \theta) / (s_d / \sqrt{n})$

KEY INSIGHT ** [4] When the prompt says "each subject" / "before and after" / "twin pairs" / "left vs right" $\rightarrow$ MATCHED PAIRS, not 2-sample.
 Subtract first, then run a 1-sample t-test on the differences. Mistaking this is a near-automatic 2-point loss.

PAGE 7 | UNITS 8-9: CHI-SQUARE · INFERENCE FOR SLOPE

CHI-SQUARE — 3 different tests, ONE formula [3]/[4]

TEST STATISTIC (all 3 chi-square tests):
 $\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$

EXPECTED COUNTS for two-way table:
 $(\text{row total} \times \text{column total}) / \text{grand total}$

CONDITIONS: random sample + ALL expected counts ≥ 5 + 10%/independent

Which Chi-Square Test Do I Use? [4]/[5]

Test	When to use	df
Goodness of Fit [4]	ONE categorical variable, ONE sample. Comparing observed counts to a CLAIMED distribution (e.g., "30/30/40 split").	df = #categories - 1
Homogeneity [4]/[5]	Compare DISTRIBUTION of one variable across SEVERAL populations (multiple SRSs from different groups).	df = (r-1)(c-1)
Independence [4]/[5]	ONE sample, TWO categorical variables. Are they associated?	df = (r-1)(c-1)

KEY INSIGHT [5] How to tell homogeneity vs independence: Look at HOW the data was collected. **Several samples** from different populations → homogeneity. **One sample** measured on two categorical variables → independence. The math is identical, the conclusion language differs.

Inference for Slope [4]/[5] — Investigative Task favorite

Population model: $\mu_y = \alpha + \beta \cdot x + \epsilon$ (β is true slope)

TEST for slope: $H_0: \beta = 0$ vs $H_a: \beta \neq 0$ (or one-sided)
 test stat: $t = b / SE_b$ df = n - 2

Read SE_b , b , t , p -value DIRECTLY from computer output table:

Predictor	Coef	SE Coef	t	P
Constant	a	SE_a	...	...
X-variable	b	SE_b	t	p

CONFIDENCE INTERVAL for β : $b \pm t^* \cdot SE_b$

WATCH OUT [4] Conditions for slope inference (LINER): Linear (residual plot random), Independent (10% rule), Normal residuals (boxplot ~symmetric), Equal SD (residual plot has no fan/cone), Random sample/assignment.

Type I, Type II Errors, Power [4]/[5]

Concept	What it means
Type I error [4]	Reject H_0 when H_0 is true. Probability = α (significance level). "Falsely conclude effect exists."
Type II error [4]	Fail to reject H_0 when H_0 is false. Probability = β . "Miss a real effect."
Power [4]/[5]	$1 - \beta = P(\text{correctly rejecting false } H_0)$. Want this HIGH.
To increase power [5]	Increase n (best); increase α (tradeoff with Type I); larger true effect; reduce variability.

PAGE 8 | THE INFERENCE DECISION TREE — WHICH TEST?

This page is the AP Stats super-tool. Most FRQ points are lost not from bad math but from picking the WRONG procedure. Use this tree.

STEP 1 — What kind of variable? [3]

CATEGORICAL (yes/no, color, brand)	→ proportions OR chi-square
QUANTITATIVE (height, time, score)	→ means OR slope

STEP 2 — How many groups? [3]/[4]

Variable type	1 group	2+ groups
Categorical, 2 levels (yes/no) [3]	1-PROP z-test / interval	2-PROP z-test / interval
Categorical, claimed distribution [4]	χ^2 Goodness of Fit	—
Categorical, 2 vars in ONE table [4]	—	χ^2 Homogeneity (multiple samples) or Independence (one sample)
Quantitative [3]	1-SAMPLE t-test / interval	2-SAMPLE t (independent) or MATCHED PAIRS t (paired)
Quantitative, relationship to another QV [4]/[5]	—	t-test for slope (regression)

STEP 3 — Test or Interval? [3]/[4]

- **Test [3]** if the prompt asks: "is there evidence that...", "is the proportion different from", "do the means differ"
- **Interval [3]** if the prompt asks: "estimate", "how confident are we", "give a range"

STEP 4 — Verify Conditions BEFORE running it [3]/[4]

Procedure	Conditions to check (and STATE on FRQ)
1-prop / 2-prop z	Random + 10% (each) + Large Counts (each $n\hat{p} \geq 10$, $n(1-\hat{p}) \geq 10$) — for CI use $\hat{p}$; for test use p_0
1-sample / 2-sample t	Random + 10% (each) + Normal (population normal OR $n \geq 30$ OR sample plot symmetric)
Matched pairs t	Random pairing + 10% (paired) + Normal ($n_d \geq 30$ OR symmetric DIFFERENCES)
χ^2 (any)	Random + 10% + ALL expected counts ≥ 5
Slope t-test	L-I-N-E-R (linear, independent, normal residuals, equal spread, random)

INVESTIGATIVE TASK (Q6) — what to expect [4]/[5]

Q6 is worth ~25% of FRQ score. Always 7-9 sub-parts. Layered: starts with familiar ideas, builds to a NEW concept you haven't seen. Common Q6 themes: simulation-based inference, novel sampling distribution, custom test statistic, two-way ANOVA-like extension.

- **[4] Strategy:** read THE WHOLE QUESTION FIRST. Subparts often refer back.
- **[4] Don't panic on the new part.** They give you the formula or distribution. Apply it like a familiar test.
- **[5] Connect to known procedures:** "This is similar to a 2-sample t-test, but using simulation instead of t-distribution because conditions aren't met."
- **[5] Always WRITE in proper statistical English** even on the novel parts. The conclusion templates apply.

PAGE 9 | FRQ STRATEGY · LANGUAGE TRAPS · CALCULATOR

FRQ POINT BREAKDOWN — what graders reward [3]/[4]/[5]

Behavior	Why it earns points
Always define parameter in context.	"Let μ = the true mean commute time of Boston commuters" — earns the parameter point.
Always state hypotheses BOTH in symbols AND words.	Symbols alone or words alone often loses the H point.
Verify each condition with a SPECIFIC justification.	"Random: SRS stated" or "Large counts: $n\hat{p} = 80 \geq 10$ " — vague "conditions met" earns 0.
Show the test name + the formula or calculator command.	"2-sample t-test, $t = \dots$ " — graders need to see what you did.
Conclusion ALWAYS in context with proper template.	Reject/fail to reject + p-value comparison + words about H_a in the actual scenario.
Use "convincing evidence" — never "prove".	Statistics never proves; it provides evidence.

LANGUAGE TRAPS — single-word point losses [3]/[4]

- "Accept H_0 " → 0 points on conclusion. Use "fail to reject".
- "Probability the parameter is in interval" [4] → WRONG. The parameter is fixed. Use "we are X% confident the true parameter is between..."
- "This proves..." → WRONG. Statistics provides evidence; never proves.
- "Causation" from observational data [4] → can never claim. Only experiments with random assignment establish cause.
- Forgetting "true" in interval interpretations [4] → "...the mean is between" misses point. "...the **true** mean of [population] is between" earns it.
- "Significant" without "statistically" [4] — be precise.

CALCULATOR ESSENTIALS [3]/[4]

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STAT > TESTS:
1: Z-Test          5: 1-PropZTest      8: TInterval
2: T-Test          6: 2-PropZTest      9: 2-SampZInt
3: 2-SampZTest     7: ZInterval         0: 2-SampTInt
4: 2-SampTTest     A: 1-PropZInt       B: 2-PropZInt
C:  $\chi^2$ -Test (matrix-based, for homogeneity/independence)
D:  $\chi^2$  GOF-Test  E: 2-SampFTest      F: LinRegTTest
G: LinRegTInt

DISTRIBUTIONS (2nd VARS):
normalcdf(low, high,  $\mu$ ,  $\sigma$ )      tcdf(low, high, df)
invNorm(area,  $\mu$ ,  $\sigma$ )               $\chi^2$ cdf(low, high, df)
binompdf(n, p, x)    binomcdf(n, p, x)
geometpdf(p, x)      geometcdf(p, x)

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WATCH OUT [4] Calculator output alone earns **NO** points if not paired with proper notation, hypotheses, conditions, conclusion. Show your test name + result.

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: 22-26 / 40	MCQ: 27-32 / 40	MCQ: 33-38 / 40
Pick ANY correct test, write hypotheses, get a p-value.	Pick the RIGHT test, verify all conditions, conclude in context.	Right test, conditions, AND complex understanding (errors, power, robustness).
Use SOCS to describe distributions in context.	Distinguish data dist. vs sampling dist.	Investigative Q6: connect novel idea to known framework.
1-prop z + 1-sample t — be solid here.	2-prop, 2-sample t, matched pairs — know which prompt = which.	All inference + chi-square + slope mastery.
Conclusion templates memorized.	Plus: confidence interval interpretation correct.	Plus: power, Type I/II tradeoffs, study-design critiques.

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:

- ✓ MCQ: 22+ correct.
- ✓ Use SOCS for every "describe distribution" — and name the variable.
- ✓ Write hypotheses in symbols AND words.
- ✓ Always pair p-value with α , then "reject" or "fail to reject".
- ✓ Conclude in context with the actual scenario.

For a 4 — add these on top:

- ✓ MCQ: 27+ correct.
- ✓ Verify ALL conditions specifically (not vague).
- ✓ Distinguish 2-sample t from MATCHED PAIRS — the give-away words.
- ✓ Distinguish observational study (no causation) from experiment (yes causation).
- ✓ Slope inference: read coefs/SE/p-value from output table.
- ✓ Use "true" in CI interpretations: "...the true mean..."

For a 5 — add these on top:

- ✓ MCQ: 33+ correct.
- ✓ Q6 Investigative Task: link new ideas to familiar inference framework.
- ✓ Power, Type I, Type II — explain in context.
- ✓ Distinguish chi-square homogeneity vs independence by sampling design.
- ✓ Address robustness when conditions are borderline.
- ✓ When experiments include blocking, EXPLAIN why blocking reduces variability.

THE ONE THING THAT MATTERS MOST [3][4][5]: ALWAYS ANSWER IN CONTEXT. AP Stats rewards students who write statistical conclusions in real-world English. "Reject H_0 , $p < 0.05$ " earns half-credit. "Since $p = 0.012 < \alpha = 0.05$, we reject H_0 . There is **convincing evidence that the mean response time of the new drug is less than the old drug for hospital patients with chronic migraines**" earns full credit. Same math, full points. **Templates + context = 5.**

If you do nothing else: memorize the conclusion templates, distinguish the 3 distributions, and use the inference decision tree on every FRQ.

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